Practical:6.1

Aim: WAPT make use of static variable and static function in class.

Code:

LAB-7 > Pelindrome.java > Pelindrome > main(String[])

```java
import java.util.*;

public class Pelindrome {
    Run | Debug
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String str=sc.nextLine();
        int i=0;
        int j=str.length()-1;
        boolean isPelinDrome=true;
        while(i<j){
            if(str.charAt(i)==str.charAt(j)){
                i++;
                j--;
            }
            else{
                isPelinDrome=false;
                break;

            }
        }
        if(isPelinDrome){
            System.out.println(x: "The string is a palindrome.");
        }
        else{
            System.out.println(x: "The string is not a palindrome.");
        }
    }
}
```

Output:

```
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-6> java Practical1
Enter the String
My name is Akshita
a : 3
e : 1
h : 1
i : 2
k : 1
m : 2
n : 1
s : 2
t : 1
y : 1
```

Practical:6.2

Aim: Create a class which ask the user to enter a sentence, and it should display count of each vowel type in the sentence.

Code:

LAB-6 > StaticExample.java > StaticExample

```java
import java.util.*;

class Demo{
  static int count=0;
  Demo(){
    count++;
  }
  static void printCount(){
    System.out.println("Count: "+count);
  }
}
public class StaticExample {
  Run | Debug
  public static void main(String[] args){
    Demo s1=new Demo();
    s1.printCount();
    Demo s2=new Demo();
    s2.printCount();
    Demo s3=new Demo();
    s3.printCount();
  }
}
```

Output:

```
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-6> javac StaticExample.java
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-6> java StaticExample
Count: 1
Count: 2
Count: 3
```

Practical:6.3

Aim: The program should continue till user enters a word "quit". Display the total count of each vowel for all sentences.

Code:

```
test.java > test > main(String[])
import java.util.*;
public class test {
  Run | Debug
  public static void main(String []args){
    while(true){
      System.out.println(x: "Enter the String");
      String str=new Scanner(System.in).nextLine();
      if(str.compareToIgnoreCase(str: "quit")==0){
        break;
      }
      int vowel=0;
      String s=str.toLowerCase();
      for(int i=0;i<str.length();i++){
        if(s.charAt(i)=='a' || s.charAt(i)=='e' || s.charAt(i)
        =='i' || s.charAt(i)=='o' || s.charAt(i)=='u'){
          vowel++;
        }
      }
      System.out.println("Vowels: "+vowel);
    }
  }
}
```

Output:

```
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-6> java test
Enter the String
Hello
Vowels: 2
Enter the String
Akshita
Vowels: 3
Enter the String
Quit
```

Practical:7.1

Write a program to find that given number or string is palindrome

Code:

LAB-7 > Pelindrome.java > Pelindrome > main(String[])

```java
import java.util.*;

public class Pelindrome {
    Run | Debug
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String str=sc.nextLine();
        int i=0;
        int j=str.length()-1;
        boolean isPelinDrome=true;
        while(i<j){
            if(str.charAt(i)==str.charAt(j)){
                i++;
                j--;
            }
            else{
                isPelinDrome=false;
                break;

            }
        }
        if(isPelinDrome){
            System.out.println(x: "The string is a palindrome.");
        }
        else{
            System.out.println(x: "The string is not a palindrome.");
        }
    }
}
```

Output:

```
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-7> java Pelindrome
NAYAN
The string is a palindrome.
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-7> cd ..
```